Credit-Market Sentiment and the Business Cycle: A Replication of López-Salido et al. (2017)

Fernando Raffo Pedraza Bangjie Xu

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Abstract

We replicate the core results of López-Salido et al. (2017), who show that elevated credit-market sentiment forecasts a slowdown in real activity about two years ahead. We describe the data, present summary statistics of the underlying series, and reproduce every table and figure across the replication sample, an extended sample, and an Aaa-spread variant.

1 Overview

When credit is "frothy"—the López-Salido et al. (2017) thesis—spreads are narrow and low-quality issuance is high; those conditions mean-revert as activity turns down.

2 Data Sources

- **FRED** (Federal Reserve Bank of St. Louis): Baa/Aaa yields, Treasury yields, GDP. Several series are a newer vintage and must be stitched together across multiple FRED series; where the appendix is silent about which underlying series to use, we made documented judgement calls.
- **Shiller** (Shiller, 2015): long-run market data and the P/E10 ratio.
- **Greenwood–Hanson high-yield share** (Greenwood and Hanson, 2013), pulled from the HBS workbook (Harvard Business School Behavioral Finance and Financial Stability Project) and spliced with Mergent FISD; a newer vintage than the paper.

3 Replication Sample (1929–2015)

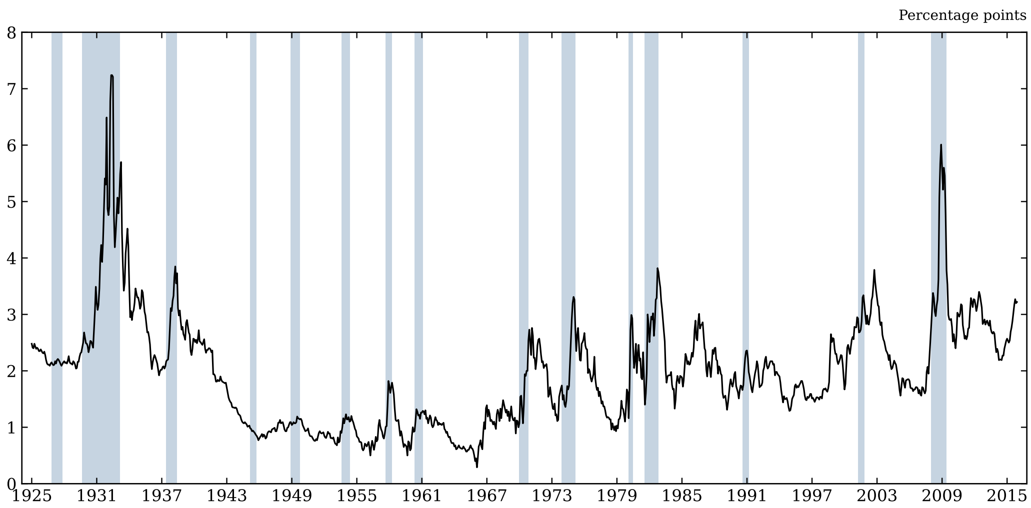

Figure 1: Figure I. Baa–Treasury credit spread with NBER recessions shaded. *Takeaway:* the spread spikes into every recession, motivating it as a cyclical signal.

Table 1: Table I. One-year-ahead GDP growth on the lagged credit-spread change vs. equity returns. *Takeaway:* the credit term is significant where equity returns are not.

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.958***	—	-2.124***
	(0.585)		(0.578)
r_t^{SP}	—	0.075**	0.022
		(0.033)	(0.035)
Δy_{t-1}	0.476***	0.478***	0.464***
	(0.104)	(0.136)	(0.114)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.248
			(0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.814**
			(0.360)
π_{t-1}	—	—	0.095
			(0.076)
$\bar{R}^2$	0.426	0.381	0.453
Standardized effect on Δy_t			
Δs_{t-1}	-0.365	—	-0.396
r_t^{SP}	—	0.299	0.087

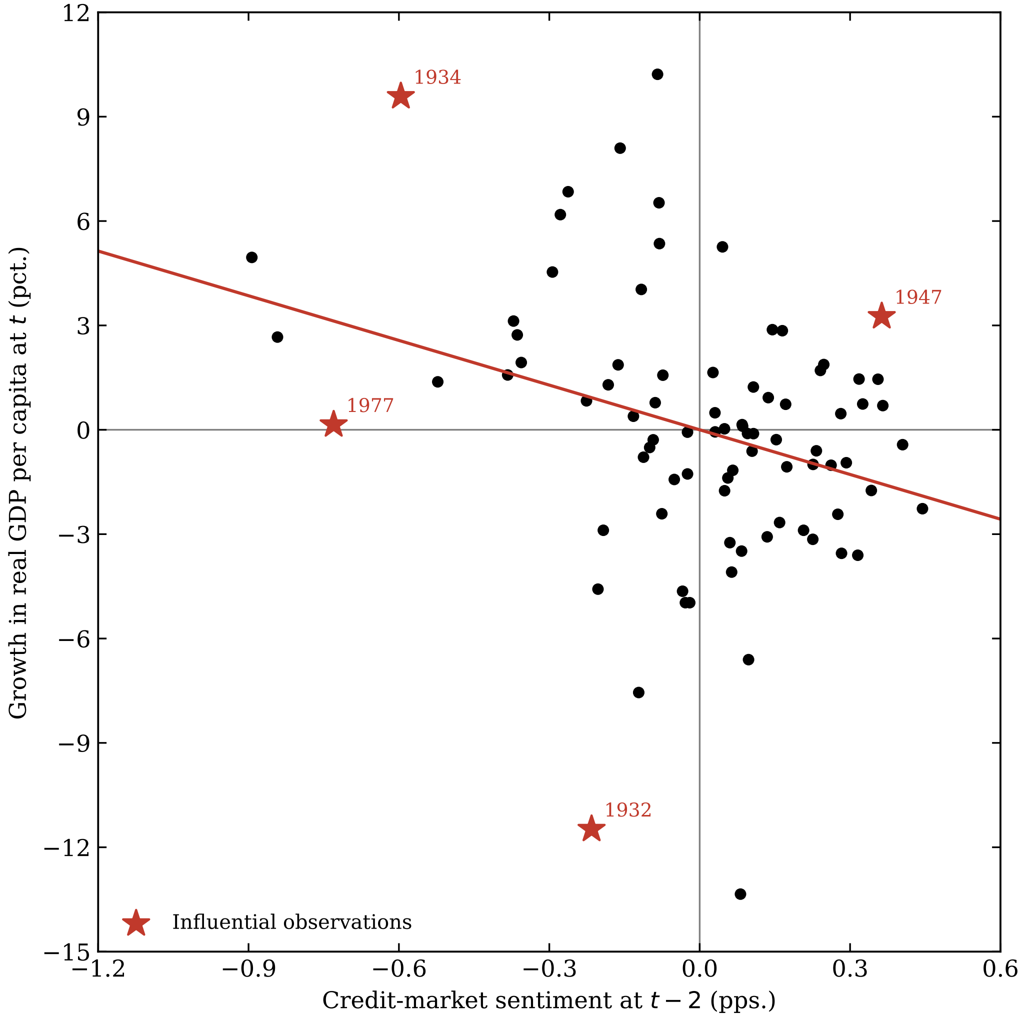

Figure 2: Figure II. Credit-market sentiment vs. subsequent growth. *Takeaway:* the downward slope shows frothy credit precedes weaker growth.

Table 2: Table II. Sentiment-based predictive regressions with the auxiliary decomposition. *Takeaway:* lagged log high-yield share and spread jointly forecast spread changes.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.281***	—	-3.861***	-4.907***
	(1.253)		(1.301)	(1.678)
$\hat{r}_t^{SP}$	—	0.145*	0.053	—
		(0.082)	(0.074)	
Δy_{t-1}	0.597***	0.534***	0.590***	0.581***
	(0.114)	(0.099)	(0.112)	(0.096)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.099
				(0.313)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.616
				(0.414)
π_{t-1}	—	—	—	0.123
				(0.154)
R^2	0.377	0.337	0.380	0.394

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \mathrm{HYS}_{t-2}$	0.147***	—
	(0.042)	
s_{t-2}	-0.245***	—
	(0.047)	
$\ln[P/E10]_{t-2}$	—	-0.132***
		(0.040)
R^2	0.102	0.083

4 Extended Sample (1929–2025)

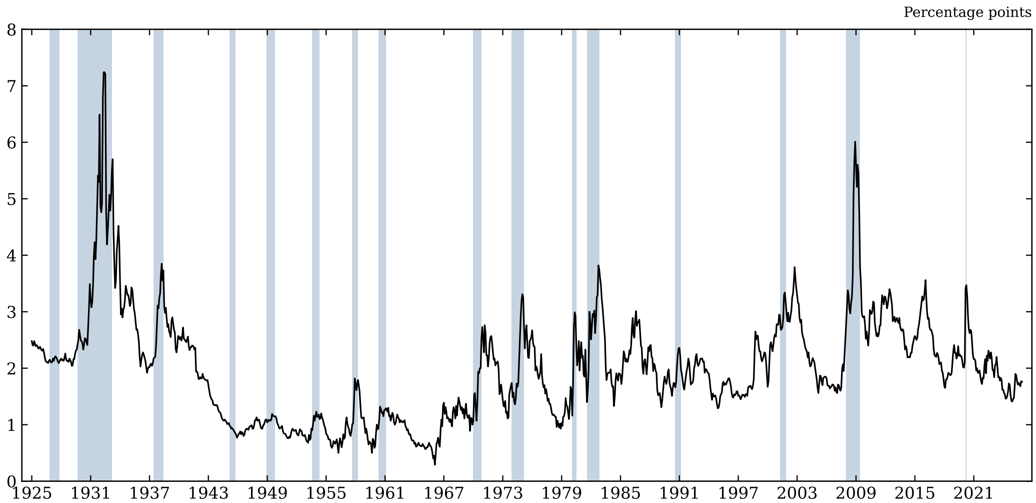

Figure 3: Figure I, extended. *Takeaway:* the cyclical pattern persists through recent data.

Table 3: Table I, extended sample. *Takeaway:* the credit result survives out of the original window.

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.871***	—	-2.033***
	(0.566)		(0.571)
r_t^{SP}	—	0.069**	0.017
		(0.032)	(0.034)
Δy_{t-1}	0.461***	0.465***	0.449***
	(0.108)	(0.136)	(0.116)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.170
			(0.265)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.776**
			(0.358)
π_{t-1}	—	—	0.089
			(0.078)
$\bar{R}^2$	0.395	0.352	0.415
Standardized effect on Δy_t			
Δs_{t-1}	-0.350	—	-0.380
r_t^{SP}	—	0.283	0.068

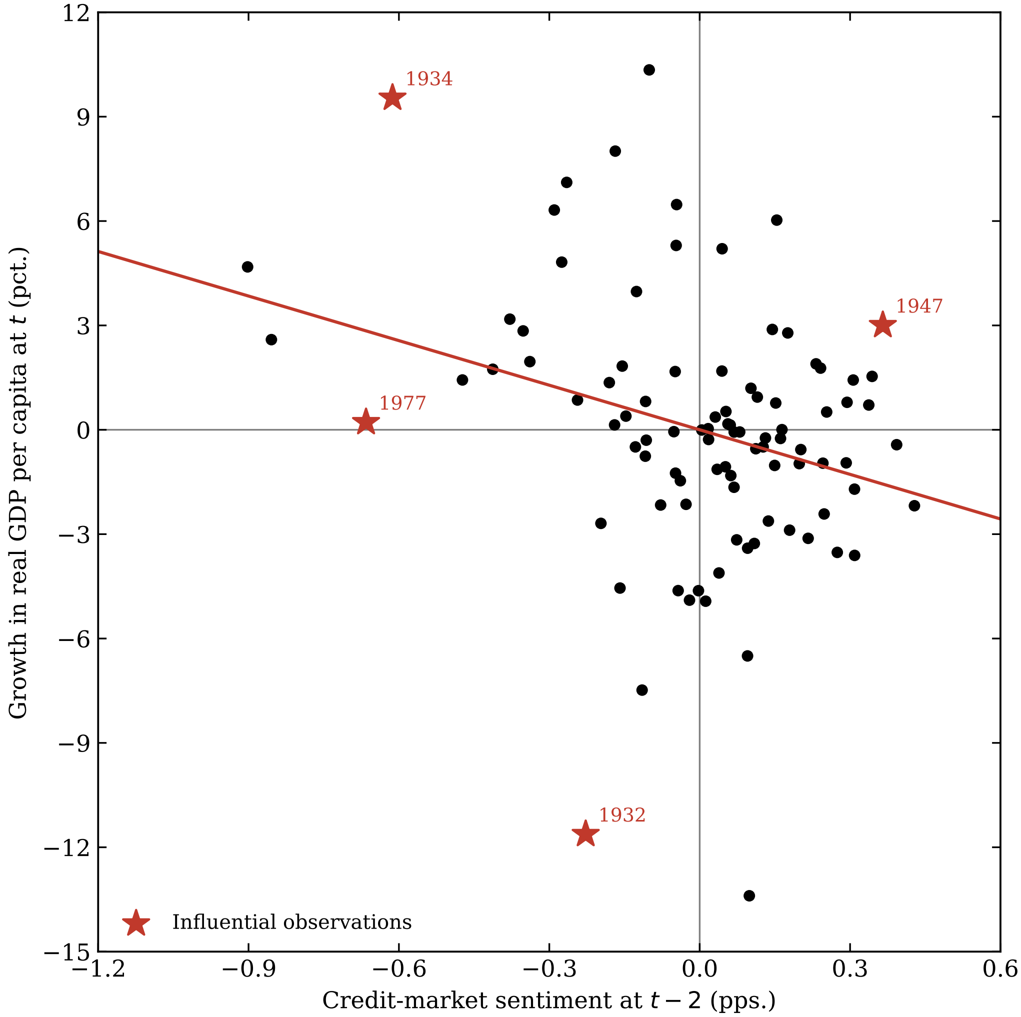

Figure 4: Figure II, extended. *Takeaway:* the negative sentiment–growth relation holds up.

Table 4: Table II, extended sample. *Takeaway:* coefficients are close to the replication values.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta\hat{s}_t$	-4.271***	—	-3.884***	-4.900***
	(1.287)		(1.307)	(1.679)
$\hat{r}_t^{SP}$	—	0.176*	0.068	—
		(0.101)	(0.089)	
Δy_{t-1}	0.577***	0.516***	0.570***	0.559***
	(0.118)	(0.103)	(0.115)	(0.099)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.117
				(0.286)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.603
				(0.402)
π_{t-1}	—	—	—	0.121
				(0.154)
R^2	0.353	0.313	0.356	0.368

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \mathrm{HYS}_{t-2}$	0.130***	—
	(0.039)	
s_{t-2}	-0.244***	—
	(0.046)	
$\ln[P/E10]_{t-2}$	—	-0.091***
		(0.035)
R^2	0.096	0.047

5 Extension: Aaa Credit Spread

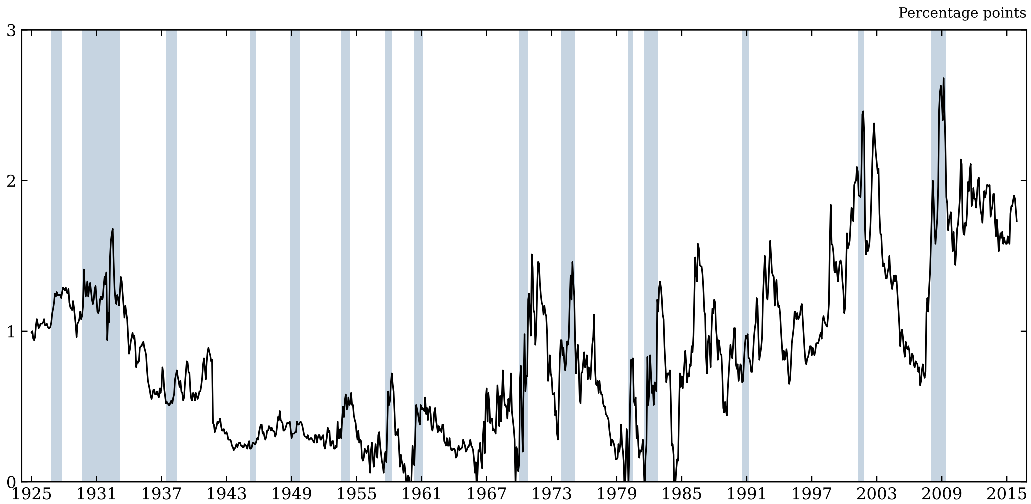

Figure 5: Figure I, Aaa (replication). *Takeaway:* the higher-grade spread is smoother but still cyclical.

Table 5: Table I, Aaa (replication). *Takeaway:* the Aaa spread carries weaker but same-signed predictive content.

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.958***	—	-3.192***
	(0.734)		(1.168)
r_t^{SP}	—	0.075**	0.056
		(0.033)	(0.035)
Δy_{t-1}	0.517***	0.478***	0.477***
	(0.109)	(0.136)	(0.121)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.326
			(0.314)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.917*
			(0.504)
π_{t-1}	—	—	0.102
			(0.108)
$\bar{R}^2$	0.321	0.381	0.396
Standardized effect on Δy_t			
Δs_{t-1}	-0.165	—	-0.269
r_t^{SP}	—	0.299	0.225

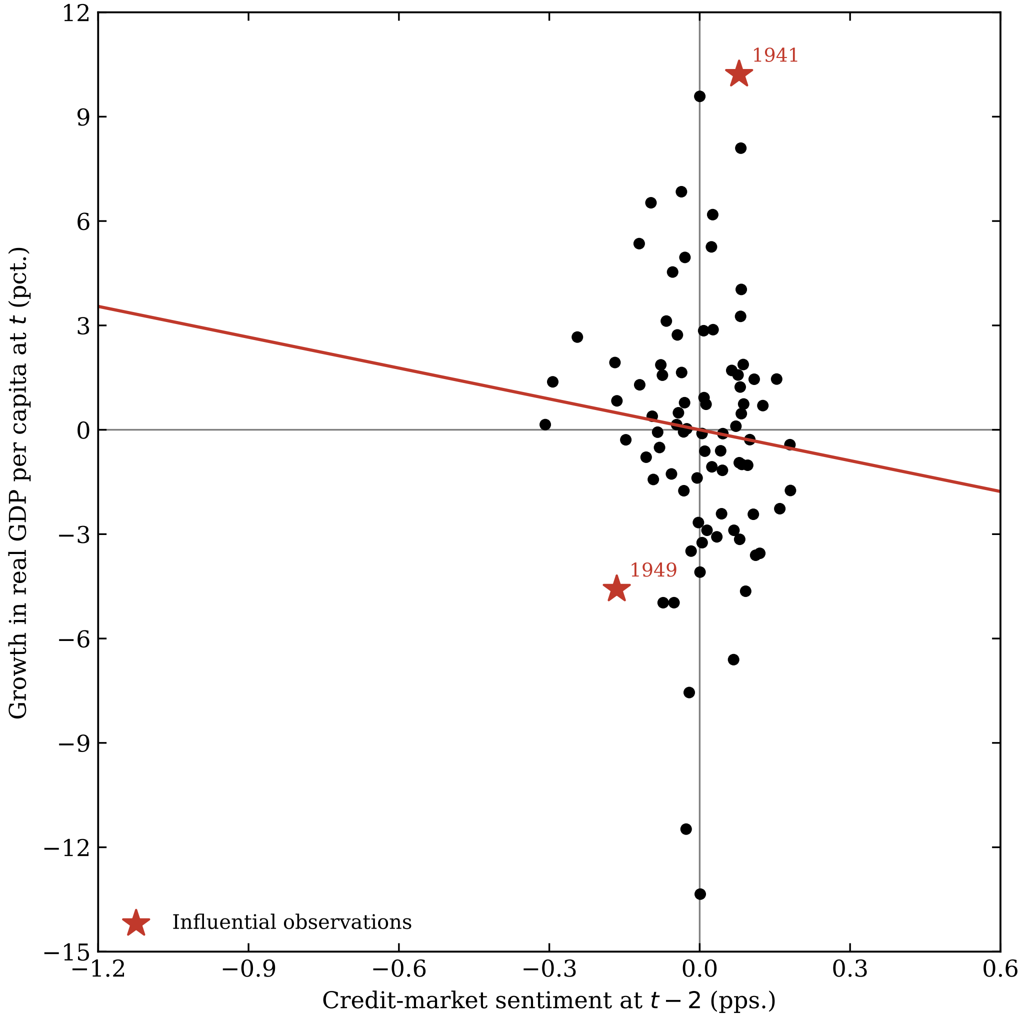

Figure 6: Figure II, Aaa (replication). *Takeaway:* sentiment built on Aaa shows a flatter slope.

Table 6: Table II, Aaa (replication). *Takeaway:* the auxiliary decomposition is weaker for the higher-grade spread.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.956	—	-2.362	-3.663
	(2.488)		(2.865)	(2.806)
$\hat{r}_t^{SP}$	—	0.145*	0.131	—
		(0.082)	(0.085)	
Δy_{t-1}	0.555***	0.534***	0.549***	0.559***
	(0.108)	(0.099)	(0.100)	(0.105)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.063
				(0.308)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.373
				(0.394)
π_{t-1}	—	—	—	0.043
				(0.132)
R^2	0.321	0.337	0.342	0.328

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \text{HYS}_{t-2}$	0.078***	—
	(0.029)	
s_{t-2}	-0.167***	—
	(0.058)	
$\ln[P/E10]_{t-2}$	—	-0.132***
		(0.040)
R^2	0.064	0.083

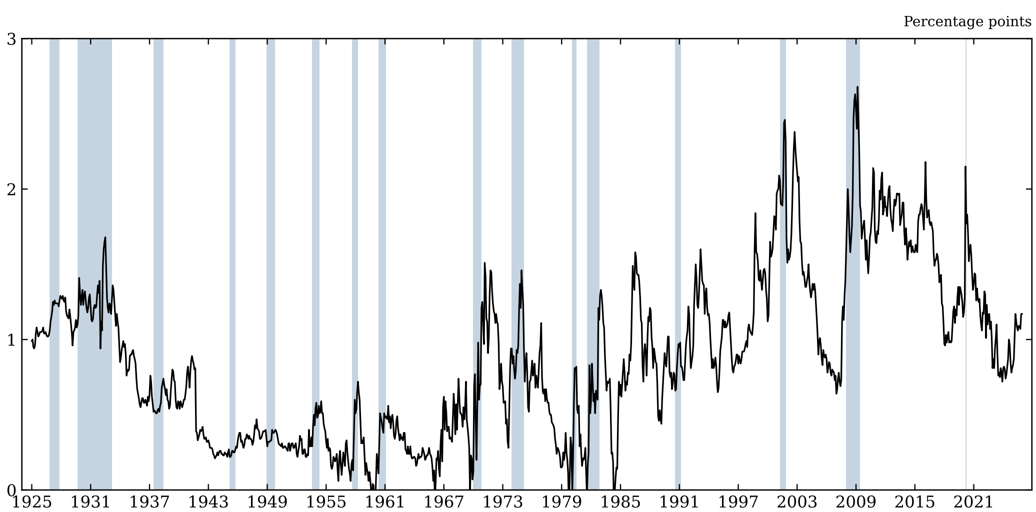

Figure 7: Figure I, Aaa (extended). *Takeaway:* the Aaa pattern extends to the present.

Table 7: Table I, Aaa (extended). *Takeaway:* extended Aaa estimates remain same-signed.

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.812**	—	-2.926**
	(0.713)		(1.142)
r_t^{SP}	—	0.069**	0.050
		(0.032)	(0.034)
Δy_{t-1}	0.500***	0.465***	0.463***
	(0.112)	(0.136)	(0.124)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.246
			(0.285)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.829*
			(0.478)
π_{t-1}	—	—	0.092
			(0.108)
$\bar{R}^2$	0.298	0.352	0.360
Standardized effect on Δy_t			
Δs_{t-1}	-0.154	—	-0.248
r_t^{SP}	—	0.283	0.206

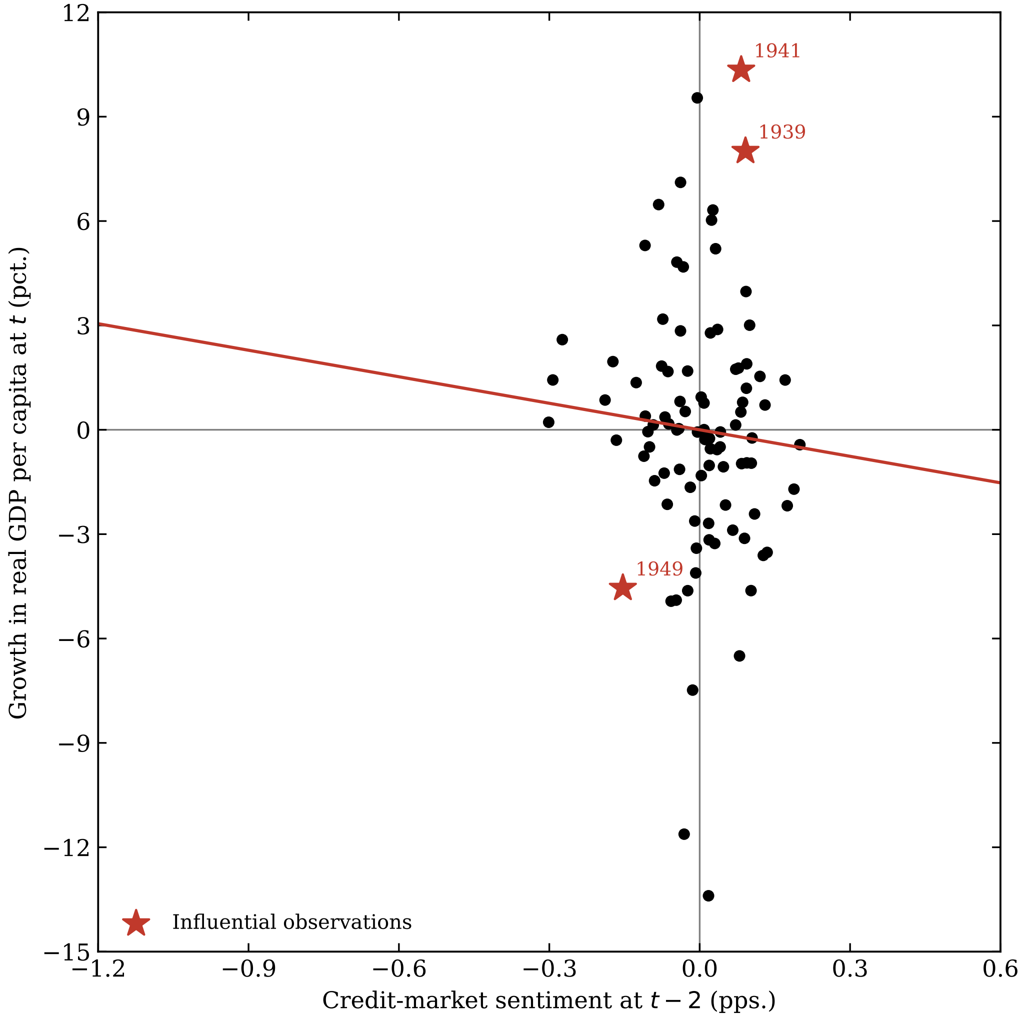

Figure 8: Figure II, Aaa (extended). *Takeaway:* the relation weakens but does not reverse.

Table 8: Table II, Aaa (extended). *Takeaway:* consistent with the replication-sample Aaa result.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.541	—	-2.489	-3.195
	(2.342)		(2.532)	(2.652)
$\hat{r}_t^{SP}$	—	0.176*	0.162	—
		(0.101)	(0.104)	
Δy_{t-1}	0.536***	0.516***	0.532***	0.540***
	(0.111)	(0.103)	(0.103)	(0.107)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.047
				(0.281)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.372
				(0.382)
π_{t-1}	—	—	—	0.041
				(0.132)
R^2	0.298	0.313	0.318	0.306

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \text{HYS}_{t-2}$	0.077***	—
	(0.028)	
s_{t-2}	-0.185***	—
	(0.058)	
$\ln[P/E10]_{t-2}$	—	-0.091***
		(0.035)
R^2	0.070	0.047

6 Case Study: The Sentiment Signal Out of Sample, 2020–2022

The post-2008 reconstruction of the high-yield issuance share from Mergent FISD makes it possible to apply the Table II first stage, estimated on 1929–2015 and held fixed, to the COVID cycle. The figure below compares the predicted change in the Baa–Treasury spread—using the full two-ingredient sentiment signal and, separately, the spread-reversion leg alone—against the realized change. The full walkthrough and its caveats are in `04_case_study.ipynb`.

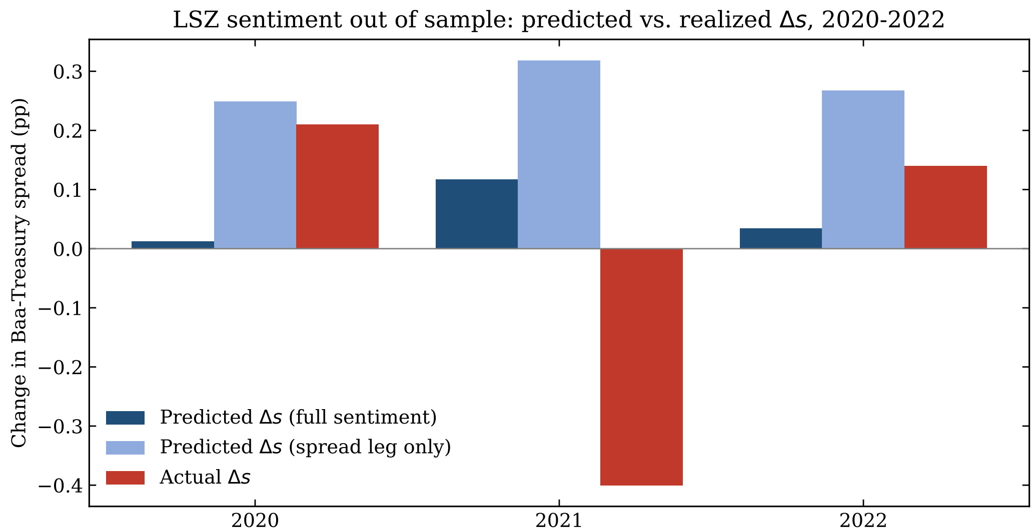

Figure 9: Out-of-sample application of the sentiment first stage to 2020–2022: predicted versus realized change in the credit spread. *Takeaway:* the froth term is only observable post-2008 thanks to the FISD reconstruction.

7 Successes and Challenges

The long-run spread reproduces cleanly and the credit-beats-equity result holds ($R^2 \approx 0.38$). The main difficulties were sourcing the high-yield share and matching the paper's Newey and West (1987) standard errors; vintage and series-stitching choices explain small differences from the published numbers.

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